

Jihao Xin

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Introduction

I'm a **PhD student in Computer Science**, working with Prof. Marco Canini at SANDS Lab, **KAUST**. I'm working on **Agentic Systems**, with a focus on **AI Scientists**, autonomous systems for scientific discovery. I built **idea2paper.org**, a *Research-as-a-Service* system that turns a research idea into a full paper. I recently started working on **AttentionHub.org**, an Agent-Native knowledge library for AI Scientists. Previously, I worked on distributed machine learning systems: model compression, KV-Cache compression, and efficient communication.

I am deeply **curious** about technology and love building products. I am drawn to the frontier and learn fast. I am a strong communicator, fluent in **Chinese and English**, professionally and in daily life.

Education

January 2023 – Present	📖 Ph.D. student in Computer Science , KAUST, Saudi Arabia. GPA = 3.87/4.0
August 2021 – December 2022	📖 Master of Science in Computer Science , KAUST, Saudi Arabia. Dean's List Award GPA = 3.88/4.0
October 2020 – October 2021	📖 Master of Science in Applied Computational Science and Engineering , Imperial College London, UK. Distinction Degree
September 2016 – June 2020	📖 Bachelor of Engineering in Computer Science and Technology , Shandong University, China. Outstanding Student Scholarship & Outstanding Student Leader Scholarship GPA = 89.34/100

Internships

September 2023 – March 2024	📖 Research Intern , Microsoft Research Asia, Beijing, China. Supervised by Dr. Changho Hwang Efficient GPU communication for distributed AI tasks.
May 2022 – August 2022	📖 Research Intern , IST Austria, Klosterneuburg, Austria. Supervised by Prof. Dan Alistarh TopK compression with efficient CUDA-based threshold selection.
August 2019 – November 2019	📖 Engineering Intern , Tencent Cloud, Shenzhen, China. Tencent international website backend development by GoLang.
June 2019 – August 2019	📖 Research Intern , The University of Western Australia, Remote. YOLO-based splash detection of Western Australia's coastal highway by Python.

Research Publications

- 1 **J. Xin** and M. Canini, "IdeaDrift: Quantifying idea drift in autonomous AI scientists via structured snapshot," in *Submission*, 2026.

- 2 **J. Xin**, T. Lyu, Q. Pan, K. Wang, and M. Canini, “Why smaller is slower? dimensional misalignment in compressed LLMs,” in *EuroMLSys*, **Oral Presentation**, Edinburgh, UK, 2026.
- 3 **J. Xin**, T. Lyu, D. Keyes, H. Ltaief, and M. Canini, “KV-Cache compression via RoPE-aligned pruning,” in *arXiv preprint arXiv:2602.02599*, 2026.
- 4 **J. Xin**, M. Canini, P. Richtárik, and S. Horváth, “Quantize once, train fast: Allreduce-compatible compression with provable guarantees,” in *ECAI*, **Oral**, Bologna, Italy, 2025.
- 5 J. Weng, B. Han, D. Gao, R. Gao, W. Zhang, A. Zhong, C. Xu, **J. Xin**, Y. Luo, L. W. Wills, and M. Canini, “Assassyn: A unified abstraction for architectural simulation and implementation,” in *ISCA*, Tokyo, Japan, 2025.
- 6 N. Xue, D. Guo, J. Zhang, **J. Xin**, Z. Li, and X. Huang, “OpenFunction for software defined IoT,” in *ISNCC*, **Best Paper Award**, Dubai, UAE, 2021.
- 7 **J. Xin**, M. Canini, P. Richtárik, and S. Horváth, “Global-QSGD: Allreduce-compatible quantization for distributed learning with theoretical guarantees,” in *EuroMLSys*, **Spotlight**, Rotterdam, Netherlands, 2025.
- 8 **J. Xin**, S. Bae, K. Park, M. Canini, and C. Hwang, “Immediate communication for distributed AI tasks,” in *HotInfra*, **Oral**, Austin, USA, 2024.
- 9 **J. Xin**, I. Ilin, S. Zhang, M. Canini, and P. Richtárik, “Kimad: Adaptive gradient compression with bandwidth awareness,” in *DistML*, Paris, France, 2023.
- 10 M. Mao, X. Feng, **J. Xin**, and T. W. S. Chow, “A convolutional neural network-based maximum power point voltage forecasting method for pavement PV array,” *IEEE TIM*, 2023.

Miscellaneous Experience

Teaching Assistant

2025 Spring	KAUST CS398 Graduate Seminar
2022 Fall, 2024 Fall, 2025 Fall	KAUST CS240: Computing Systems and Concurrency
2023 Spring	Saudi Ministry of Interior Training Program CS229: Machine Learning
2022 Summer, 2023 Summer	UESTC&KAUST Summer Camp

Seasonal Programs

2025 Summer	PUMPS+AI Summer School, Barcelona Supercomputing Center, Spain.
2019 Summer	Summer Research Program, The University of Western Australia.
2019 Winter	Robotics & AI Winter School, Imperial College London Hamlyn Centre.

Leadership

2024-2025	KAUST Graduate Student Council Representative (12/1683).
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Skills

AI Agent	Claude Code, OpenHands, Agent Loop, State Management, Memory, Context Engineering, Skills, Sandbox Execution, Long-horizon Automation
Programming Languages	C++/CUDA, Python/PyTorch, GoLang
	Chinese (Native), English (Professional)